

Ex 9.5 Q23

$$\sum \frac{2+(-1)^n}{(1.25)^n}$$

$$\begin{aligned} \textcircled{1} (\text{geom}) \quad \sum \frac{2+(-1)^n}{(1.25)^n} &= \sum \frac{2}{(1.25)^n} + \sum \left(-\frac{1}{1.25}\right)^n \\ &= \underbrace{2 \sum \left(\frac{1}{1.25}\right)^n}_{\text{Conv.}} + \underbrace{\sum \left(-\frac{1}{1.25}\right)^n}_{\text{Conv.}} \\ &\quad \underline{\left|\frac{1}{1.25}\right| < 1} \qquad \left|-\frac{1}{1.25}\right| < 1 \end{aligned}$$

$$\begin{aligned} \textcircled{2} (\text{root}) \quad \text{Let } a_n &= \frac{2+(-1)^n}{(1.25)^n} \qquad 1 < \sqrt[n]{2+(-1)^n} < 3^{\frac{1}{n}} \\ \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} &= \lim_{n \rightarrow \infty} \frac{\sqrt[n]{2+(-1)^n}}{1.25} = \frac{1}{1.25} < 1 \qquad \begin{matrix} \nearrow 1 \\ \text{(Squeeze Thm)} \end{matrix} \end{aligned}$$

246 $\sum_{n=3}^{\infty} \frac{2^{n^2}}{n^{2^n}}$

① (Ineq.)

Motivation. $\underline{n^{2^n}} \geq \underline{2^{2^n}} \geq \underline{2^{n^2}}$ for $n \geq \square$

$\frac{2^{n^2}}{n^{2^n}} \leq \boxed{1} \rightarrow$ Can we get a conv. series?

New $n^{2^n} \geq 2^{2^n} \geq 2^{n^2}$ for $n \geq \square$

$\frac{2^{n^2}}{n^{2^n}} \leq \frac{2^{n^2}}{2^{2n^2}} = \frac{1}{2^{n^2}} = \left(\frac{1}{2}\right)^{n^2} \leq \underline{\left(\frac{1}{2}\right)^n}$

$\Rightarrow$ Use DCT.

Conv. Geom. Srs.

② (Ineq.) (Equalize the power)

$$\frac{2^{n^2}}{n^{2^n}} \leq \left(\dots \right)^{n^2}$$

$$n^{2^n} \geq \underline{n}^{n^2} \geq \underline{3}^{n^2}$$

$$\frac{2^{n^2}}{n^{2^n}} \leq \frac{2^{n^2}}{3^{n^2}} = \left(\frac{2}{3} \right)^{n^2} \leq \underbrace{\left(\frac{2}{3} \right)^n}_{\text{conv.}}$$

$\Rightarrow$ DCT

$$\textcircled{3} (\text{Root}) \quad a_n = \frac{2^{n^2}}{n^{2^n}} \quad \sqrt[n]{|a_n|} = \frac{2^n}{\underline{n}^{2^n/n}} \leq \frac{2^n}{2^{2^n/n}} \\ = 2^{n - 2^n/n}$$

Claim.

$$\lim_{n \rightarrow \infty} \left(n - \frac{2^n}{n} \right) = -\infty$$

($\infty - \infty$)

~~$\frac{n^2 - 2^n}{n}$~~ (L'Hôpital) ✓

~~$\frac{n - \frac{2^n}{n}}{n} = n \left(1 - \frac{2^n}{n^2} \right)$~~

$\therefore \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \underline{\underline{0}} \quad (\text{Squeeze Thm})$

Srs Conv. by Root Test

$$a_n = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} - 1 & \text{if } n \text{ is even} \end{cases}$$

$$n = 2l + 1 \Rightarrow l = \frac{n+1}{2}$$

$$\underline{1} \quad 0 \quad \underline{2} \quad 1 \quad \underline{3} \quad 2 \quad \underline{4}$$

$$a_n = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} - 1 & \text{if } n \text{ is even} \end{cases}$$

$a_n \rightarrow \infty$ but a_n is not eventually increasing.

To check inc/dec.

① Set $a_n = \underline{f(n)}$ [Introduce $f(x)$]

② Find $\underline{f'(x)}$

Ex. 9.6 Q18

$$\sum \cancel{(-1)^n} \frac{(-1)^n}{1+\sqrt{n}}$$

$$b_n = \frac{1}{1+\sqrt{n}}$$

(AST) ① Clearly $b_n \geq 0$

② Since $\underline{1+\sqrt{n}}$ is increasing ^{on} ~~for~~ $n \geq 1$, $\Rightarrow$

b_n is decreasing on $n \geq 1$.

$$\textcircled{3} \lim_{n \rightarrow \infty} b_n = 0$$

$\therefore$ The series conv. by AST.

$$\text{Let } a_n = \frac{(-1)^n}{1+\sqrt{n}} \Rightarrow \lim_{n \rightarrow \infty} \frac{|a_n|}{\frac{1}{\sqrt{n}}} = 1 \Rightarrow \sum |a_n| \text{ ~~is~~ div.}$$

$\therefore$ Converge conditionally.